

SK REHAN AHAMED

Embedded Software Engineer

Bangalore, India | +91 8420573869 | skrehanahamed5@gmail.com | linkedin.com/in/sk-rehan-ahamed-23a4a922b | github.com/skrehanahamed

PROFESSIONAL SUMMARY

Embedded Software Engineer with 4+ years of experience in Embedded C/C++, AUTOSAR, Android, RTOS, BSP, CAN, and UDS diagnostics for automotive systems. Skilled in Python test automation, functional safety (ISO 26262, ASIL-B), and cross-platform debugging across Android, kernel, and RTOS layers. Proven track record delivering AUTOSAR BSW/RTE, instrument cluster/IVI software, and CI/CD test frameworks for Ford, Continental, Stellantis (via Marelli), and ASML.

TECHNICAL SKILLS

Programming: Embedded C, C++, C++11/14, Python, Bash, OOP, STL, DSA, Multithreading

Embedded & Automotive: AUTOSAR (MCAL, BSW, RTE, CDD), Instrument Cluster, IVI, Android, RTOS, BSP, CAN Stack, DaVinci Toolchain

Protocols & Diagnostics: CAN, CAN FD, UDS, UART, SPI, I2C, CAPL, CANoe, CANalyzer

Testing & Automation: GTest, Ceedling, Unity, Trace32, GDB, Robot Framework, Jenkins, CAN Log Analysis

Build & Dev Tools: CMake, Bazel, Docker, Git, GitHub, GitLab

GUI Development: Qt, Qt Widgets, Qt Quick, QML, Signal-Slot Programming

Functional Safety & Quality: ISO 26262, ASIL-B, TSR/SSR, FMEA, MISRA C/C++, Polyspace, Code Coverage (gcov, LCOV)

PROFESSIONAL EXPERIENCE

L&T Technology Services | Bangalore, India

Mar 2026 – Present

Senior Engineer

- Contributing to Stellantis AIDA R2 automotive software development for Marelli, working across Android, RTOS, and BSP layers on Qualcomm SA8155P and SA6155P SoCs and Renesas RH850 platforms.
- Developing and maintaining Embedded C/C++ components with Green Hills Integrity RTOS for automotive software development.
- Performing software integration, debugging, log analysis, and defect investigation across Android, kernel, and RTOS components in an Agile environment.
- Developed CAPL scripts and Robot Framework automation for CAN/UDS test cases, improving test coverage and reducing manual validation effort for instrument cluster systems.

Alten India | Bangalore, India

Nov 2025 – Mar 2026

Senior Engineer

- Migrated ASML's DART diagnostic application (semiconductor lithography equipment) from C to modern C++11/14 and transitioned the build system from CMake to Bazel.
- Contributed to software development, Python automation, debugging, fault log analysis, and Git-based source code management.

Wipro Technologies | Kolkata, India

Apr 2022 – Nov 2025

Project Engineer

- Developed AUTOSAR-based embedded software for Instrument Cluster and IVI systems (Ford, Continental) using Embedded C on Infineon TRAVEO T2G, per MISRA C and ISO 26262 ASIL-B.
- Implemented AUTOSAR application components (EPRNDL, SCPC, Headlamp) and configured MCAL, BSW, and RTE using DaVinci Developer and Configurator.
- Performed software integration, MCAL verification, CAN/CAN FD communication testing, UART debugging, and UDS diagnostics using CANoe and Trace32.
- Executed functional safety and verification activities including TSR/SSR analysis, FMEA, requirement traceability, JAMA, and TestRail management.
- Built Python automation frameworks for CAN log analysis, regression testing, and validation; developed Qt/QML HMI applications and unit tests using GTest and Ceedling.

KEY PROJECTS

Stellantis AIDA R2 – Marelli

Aug 2026 – Present

- BSP Developer on the vehicle domain side of the Green Hills Integrity kernel (Renesas RH850 MCU), alongside Android kernel/BSP on Qualcomm SA8155P/SA6155P SoCs; performing integration, debugging, and log analysis.

Suzuki Instrument Cluster Automation & Validation – Marelli

Mar 2026 – Jul 2026

- Built CAPL and Robot Framework automation for CAN/UDS diagnostics on TRAVEO T2G-based instrument clusters, using CANoe/CANalyzer for regression testing and reporting.

DART Diagnostic Tool Modernization – ASML (Offshore)

Nov 2025 – Mar 2026

- Migrated the diagnostic tool for ASML's semiconductor lithography systems from C to C++11/14 and CMake to Bazel, with Python automation for fault log analysis.

AUTOSAR Instrument Cluster & IVI Development – Ford & Continental (Offshore)

Jun 2022 – Oct 2025

- Developed AUTOSAR-based embedded software and Qt/QML HMI applications with CAN/UDS diagnostics compliant with MISRA C and ISO 26262 ASIL-B.

EDUCATION

M.Tech in Computer Science & Engineering

Aug 2022

MAKAUT, Kolkata, India | GPA: 9.10 / 10.0

B.Tech in Computer Science & Engineering

Aug 2020

Aliah University, Kolkata, India | GPA: 8.23 / 10.0

ACHIEVEMENTS

- Victory League Award (13 times) – Wipro Employee Recognition, India, 2022–2025
- Habit Flagbearer – Always Communicating – Wipro Employee Recognition, India, 2024
- Habit Flagbearer – Being Responsive – Wipro Employee Recognition, India, 2023